ViSAQA: Benchmarking Evidence Sufficiency and Scope Awareness in Vietnamese Student Affairs Question Answering

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Abstract. Student Affairs question answering requires both access to institutional evidence and the ability to withhold unsupported answers. We evaluate five sequence-to-sequence models on a revised Vietnamese benchmark of 15,879 samples across 21 topics. Two splitting protocols, three retrieval configurations, and three training seeds yield 120 evaluation combinations. We distinguish retrieval quality, answer generation on answerable questions, and rejection behavior. On the topic-disjoint test, dense retrieval with reranking achieves 74.19% Hit@3. Its mBART configuration improves answerable ROUGE-L from 12.62 to 16.26, a gain that reverses on the random split, whereas ViT5 exhibits severe over-rejection. An always-insufficient baseline attains 34.89% aggregate Exact Match, illustrating the influence of repeated rejection targets, and no configuration exceeds its binary rejection F1. These results support model-dependent benefits of retrieval and the need for separate answerability evaluation, rather than uniform gains from adding context.

Keywords: Vietnamese question answering · Retrieval-augmented generation · Abstention · Student Affairs · Benchmark evaluation

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1 Introduction

Student Affairs services at Ho Chi Minh City University of Industry and Trade (HUIT) involve regulations, notices, decisions, and procedures concerning learners' rights and obligations. Relevant evidence is distributed across institutional documents. A useful question answering (QA) system must retrieve appropriate information, generate a supported answer, and recognize when it cannot answer within its defined scope. Retrieval-augmented generation (RAG) combines external evidence with a generative model [6], but retrieving relevant text does not by itself guarantee a correct answer or a reliable rejection decision.

These requirements create an evaluation problem. A test collection may contain many unanswerable questions assigned the same reference rejection sentence. A model can then obtain substantial aggregate lexical scores by repeatedly emitting that sentence, while failing on questions that have an answer. Conversely, exact matching may penalize valid paraphrases or expose output-format failures. Evaluation should therefore separate retrieval, answer generation, and rejection instead of relying on a single whole-test ranking.

We study this problem using a revised Vietnamese Student Affairs benchmark with source-linked contexts, ordinary answers, and two explicit rejection labels. Our experiments address three questions: (RQ1) how well do alternative retrievers recover reference evidence before and after reranking; (RQ2) how do QA configurations trade answer quality against rejection on answerable questions; and (RQ3) how do observed results differ between topic-disjoint and random splits? The third question is descriptive: the two protocols have different test questions and label proportions.

The contributions are an updated 15,879-sample evaluation resource spanning 21 topics; an evaluation across five generators, three retrieval configurations, two protocols, and three seeds; and a diagnostic analysis that reports answerable-only metrics, rejection precision and recall, constant-output baselines, and decoding-related limitations. We evaluate complete QA configurations rather than claim a new retrieval architecture or a causal estimate of the isolated effect of context.

2 Related Work

Vietnamese educational QA has been investigated through intent classification and machine reading comprehension [22], language models with educational data [9], and retriever–reader–generator systems for higher-education regulations [4]. Work on Vietnamese retrieval also compares lexical and neural methods across domains [10]. These studies motivate examining retrieval and generation as separate parts of an institutional QA system. Unanswerable-question evaluation has a longer history in SQuAD 2.0 [16] and cross-domain QA [19]. Agent-based systems address ambiguous and unanswerable requests [11], while RAG benchmarks include negative rejection when available evidence is insufficient [2]. Selective QA takes a different route, abstaining when a calibrated confidence

score falls below a threshold rather than generating a rejection sentence [5]. In our setting, rejection is generated as text by the same sequence-to-sequence model that produces ordinary answers. This makes its evaluation sensitive to both the distribution of rejection targets and the rule used to interpret generated text. Our retrieval configurations build on lexical BM25 scoring [18], multilingual E5 embeddings [23], a BGE-M3-based cross-encoder reranker [1], and reciprocal rank fusion [3]. For long-form answers, model-based judging has been proposed as a complement to lexical metrics [26], but it is not used here. The focus of our study is the resulting empirical behavior on Vietnamese Student Affairs questions.

3 Dataset and Evaluation Protocols

3.1 Construction and Current Release

The benchmark originates from student survey questions and HUIT policy and procedural documents. Its construction process filters question intent, separates multiple intents, standardizes topics, and uses human review together with language-model assistance for context selection and answer drafting. Each QA record associates a question with a reference context, answer, and context source. Source identifiers support traceability and are distinct from the text supplied to the generator. The source collection used for retrieval contains 83 documents after deduplication, segmented into 2,571 indexed chunks. Chunking used 512 tokens per chunk with an 80-token overlap, corresponding to a stride of 432 tokens. The benchmark used in this study contains 15,879 records across 21 topics: 8,892 answerable questions (56.00%), 5,495 insufficient-information questions (34.61%), and 1,492 out-of-scope (OOS) questions (9.40%). All experiments and analyses reported in this paper use this 15,879-record version. Insufficient-information and OOS examples use the respective supervised responses:

> "Không có đủ thông tin trong tài liệu được cung cấp để trả lời câu hỏi này."

> "Câu hỏi nằm ngoài phạm vi hệ thống RAG nghiệp vụ công tác sinh viên."

Their meanings are, respectively, insufficient information in the provided documents and a question outside the scope of the Student Affairs RAG system. The current labels do not separately identify outdated-document and conflicting-document cases. We therefore do not report a distinct conflict-detection capability.

Label definitions and quality control. A question is labeled *insufficient information* when it lies within the Student Affairs service scope but the indexed source collection does not contain the information needed to answer it. It is labeled *OOS* when the request lies outside the services the system is defined to cover; under this boundary, for example, every Study Abroad and Student Exchange question is OOS. Labels are assigned per question from these criteria, not from topic names. Language-model drafts of answers and context selections were reviewed by humans before inclusion. Inter-annotator agreement and human-edit

rates were not measured for this release, so label reliability cannot be quantified; we list this as a limitation (Section 7).

3.2 Splits and Distributional Differences

Table 1 reports the composition of all 21 topics rather than corpus totals alone. Panel A lists, for each topic, its answerability counts, its topic-disjoint split assignment, and its counts in the random train, dev, and test splits; Panel B summarizes the split totals and the number of topics each split covers. Both protocols allocate 12,703 examples to training, 1,588 to dev (validation), and 1,588 to testing.

The topic-disjoint protocol assigns whole topics to splits (split seed 42), so each topic contributes its entire N to one split and nothing to the other two. The resulting partition has 12 training topics, four dev topics (Facilities and Learning Environment, Tuition Fees, On-campus and Off-campus Residence, and Student Civic Education), and five test topics (Academic Advising, Study Abroad and Student Exchange, Legal Education, Student Affairs Regulations, and Educational Benefits), with no topic shared across splits. The random protocol instead places every topic in all three splits, dividing each topic approximately 80/10/10 with integer rounding. This allocation does not preserve each topic's OOS proportion exactly; for example, Military Service is 3.56% OOS in train, 7.14% in dev, and 11.90% in test.

Table 1: Detailed dataset composition. Panel A: all 21 topics under both protocols. Each data cell reports N / A / I / O, followed by the OOS percentage (using that cell's N as denominator) in parentheses. A = answerable; I = insufficient information; O = out of scope; Dev = validation. "All" is the full topic, which the topic-disjoint protocol assigns wholly to one split.

		Topic-disjoint protocol		Random protocol		
ID	Topic	All	Split	Train	Dev	Test
1	School security, safety and information protection	360/217/142/1 (0.28%)	Train	288/169/118/1 (0.35%)	36/22/14/0 (0.00%)	36/26/10/0 (0.00%)
2	Insurance	706/402/304/0 (0.00%)	Train	565/322/243/0 (0.00%)	70/39/31/0 (0.00%)	71/41/30/0 (0.00%)
3	Facilities and learning environment	196/0/11/185 (94.39%)	Dev	157/0/9/148 (94.27%)	20/0/2/18 (90.00%)	19/0/0/19 (100.00%)
4	Academic advising	56/32/20/4 (7.14%)	Test	45/23/18/4 (8.89%)	6/5/1/0 (0.00%)	5/4/1/0 (0.00%)
5	Study abroad and student exchange	219/0/0/219 (100.00%)	Test	175/0/0/175 (100.00%)	22/0/0/22 (100.00%)	22/0/0/22 (100.00%)
6	Political, ideological, ethical and lifestyle education; student activities	1443/516/918/9 (0.62%)	Train	1155/412/737/6 (0.52%)	144/42/101/1 (0.69%)	144/62/80/2 (1.39%)
7	Vocational education and social work	1037/437/596/4 (0.39%)	Train	829/339/486/4 (0.48%)	104/50/54/0 (0.00%)	104/48/56/0 (0.00%)
8	Legal education	176/116/60/0 (0.00%)	Test	141/98/43/0 (0.00%)	18/10/8/0 (0.00%)	17/8/9/0 (0.00%)
9	Career guidance and employment	797/321/472/4 (0.50%)	Train	637/260/373/4 (0.63%)	80/28/52/0 (0.00%)	80/33/47/0 (0.00%)
10	Scholarships and awards	3367/2383/979/5 (0.15%)	Train	2693/1914/774/5 (0.19%)	337/232/105/0 (0.00%)	337/237/100/0 (0.00%)
11	Tuition fees	689/552/129/8 (1.16%)	Dev	551/441/107/3 (0.54%)	69/56/11/2 (2.90%)	69/55/11/3 (4.35%)

Table 1 (continued): Panel A

		Topic-disjoint protocol		Random protocol		
ID	Topic	All	Split	Train	Dev	Test
12	Military service	421/219/182/20 (4.75%)	Train	337/171/154/12 (3.56%)	42/25/14/3 (7.14%)	42/23/14/5 (11.90%)
13	On-campus and off-campus residence	124/54/70/0 (0.00%)	Dev	100/41/59/0 (0.00%)	12/4/8/0 (0.00%)	12/9/3/0 (0.00%)
14	Student Affairs regulations	1118/638/471/9 (0.81%)	Test	894/509/378/7 (0.78%)	112/64/46/2 (1.79%)	112/65/47/0 (0.00%)
15	Academic regulations	2576/1105/449/1022 (39.67%)	Train	2060/901/356/803 (38.98%)	258/96/48/114 (44.19%)	258/108/45/105 (40.70%)
16	Student support fund	786/623/163/0 (0.00%)	Train	629/490/139/0 (0.00%)	78/68/10/0 (0.00%)	79/65/14/0 (0.00%)
17	Student civic education	579/412/167/0 (0.00%)	Dev	463/325/138/0 (0.00%)	58/44/14/0 (0.00%)	58/43/15/0 (0.00%)
18	Student loans	226/180/45/1 (0.44%)	Train	181/144/36/1 (0.55%)	22/16/6/0 (0.00%)	23/20/3/0 (0.00%)
19	School health services	345/201/143/1 (0.29%)	Train	277/162/115/0 (0.00%)	34/21/13/0 (0.00%)	34/18/15/1 (2.94%)
20	Student conduct assessment	639/468/171/0 (0.00%)	Train	511/372/139/0 (0.00%)	64/50/14/0 (0.00%)	64/46/18/0 (0.00%)
21	Educational benefits	19/16/3/0 (0.00%)	Test	15/12/3/0 (0.00%)	2/2/0/0 (0.00%)	2/2/0/0 (0.00%)
	Total	15879/8892/5495/1492 (9.40%)	–	12703/7105/4425/1173 (9.23%)	1588/874/552/162 (10.20%)	1588/913/518/157 (9.89%)

Table 1: Detailed dataset composition (continued). Panel B: split totals and topic coverage. OOS percentages are within each split.

Protocol	Split	Topics	N	A	I	O	O%
Topic	Train	12	12703	7072	4564	1067	8.40
Topic	Dev	4	1588	1018	377	193	12.15
Topic	Test	5	1588	802	554	232	14.61
Random	Train	21	12703	7105	4425	1173	9.23
Random	Dev	21	1588	874	552	162	10.20
Random	Test	21	1588	913	518	157	9.89

The two protocols differ in class composition as well as in topic exposure. The OOS rate is 8.40% (1,067), 12.15% (193), and 14.61% (232) in the topic train, dev, and test splits, compared with 9.23% (1,173), 10.20% (162), and 9.89% (157) under random splitting; each rate uses all samples in the split as its denominator. Counting insufficient-information and OOS questions together, 786 of the 1,588 topic-test questions (49.50%) are unanswerable, against 675 (42.51%) in the random test. A metric difference between the protocols therefore combines changes in question content, topic exposure, and class composition, and should not be read as a controlled causal effect of topic separation.

OOS examples are also concentrated in a few topics. In the topic test, Student Affairs Regulations accounts for 1,118 of the 1,588 questions (70.40%),

while Study Abroad and Student Exchange, whose 219 questions are all labeled OOS, supplies 94.40% of the 232 OOS test examples. Likewise, Facilities and Learning Environment is 94.39% OOS and supplies 185 of the 193 OOS dev examples (95.85%). Topic and answerability are thus strongly associated in the topic-disjoint partition: OOS performance on the topic test largely reflects a single held-out topic, and generalization to other kinds of out-of-scope questions remains an open question. Across the full collection, Academic Regulations contributes 1,022 of the 1,492 OOS examples (68.50%), although only 39.67% of its own questions are OOS. Under the topic protocol it lies entirely in training, where it supplies 95.78% of the 1,067 OOS training examples; under the random protocol it still supplies 105 of the 157 OOS test examples (66.88%). OOS coverage therefore differs between protocols in topic composition, not only in frequency. Because topic and label are correlated, answerability is assigned to each question from the scope criteria rather than inferred from its topic name.

4 Retrieval and Generation Configurations

4.1 Retrieval

We compare BM25 [18], dense retrieval, and a hybrid configuration. Dense retrieval uses `intfloat/multilingual-e5-base` [23] as the encoder and `BAAI/bge-reranker-v2-m3` [1] as the reranker. The candidate budget is 20, the reranked selection parameter is three, and the hybrid reciprocal-rank-fusion [3] constant is 60. Reference contexts are used in context-conditioned training, whereas the three RAG test conditions are associated with their respective retrieval runs.

4.2 Generators and Comparison Scope

We evaluate ViT5-base [13], BARTpho-syllable-base [21], mBART-large-50-many-to-many-mmt [20,8], mT5-base, and mT5-small [24]. For each model, protocol, and seed, one training configuration receives the question alone (Q-only), and another receives the question with its reference context. The latter is evaluated with BM25, dense, or hybrid retrieval. This gives four inference conditions: Q-only, BM25, Dense, and Hybrid. The three retrieval conditions reuse the context-trained model for the corresponding seed. This yields $2 \times 5 \times 3 \times 4 = 120$ evaluation combinations; these are not 120 independently trained models. Q-only versus RAG changes both training inputs and inference inputs. It therefore compares complete configurations. Comparisons among BM25, Dense, and Hybrid hold the context-trained generator fixed within a model and seed, but there is no reported gold-context inference condition that would quantify an oracle-context gap. This condition was not run in the current experiment grid, which was designed around deployable retrieval configurations. As a result, the present results cannot attribute a given failure to retrieval rather than generation. The context-trained generators also face a train–inference mismatch: they are trained on reference contexts but tested on the top-3 retrieved chunks.

5 Experimental Setup and Metrics

5.1 Settings

Training uses seeds 22, 42, and 202, five epochs, a learning rate of 3×10^{-5}, and an effective batch size of 256. The maximum input and target lengths are 1,024 and 256 tokens, respectively. Test configurations use a maximum generation length of 256 tokens and greedy decoding (beam size 1). Inputs longer than 1,024 tokens are truncated. Because the tokenizers differ, the same input can have very different lengths: in the five cases of Section 6.4, mT5 inputs reach 1,262–1,542 tokens and are truncated, whereas the other models' inputs stay below the limit (785–871 tokens). Truncation rates were not tabulated for the full test sets, and near-duplicate questions across splits were not filtered. All experiments were conducted on a Google Cloud G4 (`g4-standard-48`) instance equipped with 48 vCPUs, 180 GB of system RAM, and a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU with 96 GB of GDDR7 memory.

Reproducibility resources. We provide a lightweight repository with selected utilities for data processing, retrieval, training, inference, and evaluation. It is intended to facilitate reproducibility rather than serve as a complete archival implementation of all experiments reported in this paper. Each benchmark record contains a question identifier, the question, the reference answer and context, topic and subtopic, source-document identifiers, the answer type, the OOS flag, and split assignments; the 2,571 indexed chunks are distributed with their document identifiers. The benchmark and utilities are intended for research use, and the license and access conditions are stated in the repository, which is publicly available at `https://github.com/chaubaonhan/ViSAQA`.

5.2 Answer Generation and Retrieval

We report Exact Match (EM), token-F1, ROUGE [7], BLEU [14,12], and BERTScore [25] statistics. EM follows exact agreement after the evaluator's normalization; it is not a human judgment of semantic correctness [17]. Scores on answerable questions are the primary generation results, while whole-test metrics are supplementary diagnostics. EM, token-F1, ROUGE, and BERTScore values stored on a zero-to-one scale are multiplied by 100 for presentation; BLEU is already on a zero-to-100 scale. Standard deviations are computed over the three training seeds and reflect optimization variability only, not sampling variability of the test questions. An SD of 0.00 means identical scores across seeds; for EM this occurs when the same number of questions is matched exactly in every seed (e.g., 11 of 802 for mBART Q-only) or when none is matched. Let G_q be the set of reference-relevant items for query q, and $R_{q,k}$ the first k retrieved items. Recall@k averages $|G_q \cap R_{q,k}|/|G_q|$, while Hit@k averages the indicator that this intersection is nonempty. They are not interchangeable when a question has multiple relevant items. MRR@k averages the reciprocal rank of the first relevant item, or zero if none is found. Retrieval is evaluated on

questions with available relevance judgments: 802 in the topic test and 912 in the random test. The latter leaves one answerable random-test question outside the reported retrieval evaluation.

5.3 Rejection and Constant Baselines

We compute binary rejection precision, recall, and F1 from the per-example output categories, treating insufficient-information and OOS cases as positive. We also report OOS F1 and the false-rejection rate (FR): the proportion of answerable questions assigned a rejection category. Formally, rejection precision is $TP/(TP + FP)$, recall is $TP/(TP + FN)$, and F1 is $2TP/(2TP + FP + FN)$. Undefined precision is set to zero. These statistics depend on the rule used to map generated outputs to rejection categories and should therefore be interpreted together with its limitations. Two deterministic baselines always emit the insufficient-information sentence or always emit the OOS sentence. On the topic test, their respective EM scores are 34.89 and 14.61; on the random test, they are 32.62 and 9.89. Both have binary rejection recall of 100%, but this alone does not indicate useful QA behavior. The baseline exposes the importance of repeated target sentences when interpreting aggregate EM.

Output-category parser. Each prediction is mapped to one of three categories. After the evaluator's normalization, a prediction is assigned insufficient information or OOS only when it matches the corresponding rejection sentence at the string level; all other outputs, including partial or paraphrased rejections, are assigned answerable. The parser does not search for a rejection sentence inside a longer output. It has not been validated against human category judgments, so its accuracy is unmeasured.

mT5 sentinel artifact. Every mT5-base and mT5-small output, in all conditions and seeds, begins with the sentinel token `<extra_id_0>`, which is retained in the decoded strings used for evaluation. This is consistent with mT5's span-corruption pretraining, in which every target begins with a sentinel [15], and suggests that five epochs at a learning rate of 3×10^{-5} may not have fully adapted the mT5 decoders. Because of the sentinel prefix, the parser assigns every mT5 output to the answerable category. The zero EM and zero rejection scores of both mT5 models are therefore artifacts of output format combined with the parser rule, not evidence that the models never emit rejection text: on the topic test, our sentinel audit flags 38.4–40.2% of mT5-base outputs under the three retrieval conditions as insufficient-information responses preceded by the sentinel, compared with 49.50% gold rejections, whereas mT5-small produces none. We do not re-score outputs after removing the sentinel, and we have not tested whether suppressing sentinel tokens during decoding or a higher learning rate removes the artifact.

6 Results and Discussion

6.1 Retrieval Quality

Table 2 separates candidate retrieval from reranking. On the topic test, dense Hit@1 increases from 45.51 to 48.38 after reranking, while Hit@3 increases from 69.08 to 74.19. Reranked dense retrieval exceeds BM25 and hybrid on the displayed topic-test metrics. Hybrid is slightly stronger than dense on the displayed random-test metrics, including Hit@3 of 74.23 versus 74.12. These differences are descriptive and are not presented as statistical-significance claims. For proportions near 74% on 802 or 912 queries, a 95% Wilson interval spans roughly ±3 points (71.05–77.10 for dense topic-test Hit@3). Differences smaller than this, such as the 0.11-point hybrid–dense gap on the random test, should not be read as rankings.

Table 2. Retrieval before and after reranking (percent).

Protocol	Retriever	Rerank	Hit@1	Recall@3	Hit@3	MRR@3
Topic	BM25	Before	26.68	42.17	47.13	35.29
Topic	BM25	After	41.77	58.67	63.84	51.06
Topic	Dense	Before	45.51	60.83	69.08	55.24
Topic	Dense	After	48.38	65.12	74.19	59.23
Topic	Hybrid	Before	38.53	55.35	65.46	50.42
Topic	Hybrid	After	47.01	63.95	72.44	57.73
Random	BM25	Before	44.08	54.46	60.31	51.26
Random	BM25	After	56.91	65.38	72.15	63.51
Random	Dense	Before	54.06	63.29	69.85	60.87
Random	Dense	After	58.77	66.93	74.12	65.42
Random	Hybrid	Before	55.04	62.47	69.52	61.46
Random	Hybrid	After	58.99	66.91	74.23	65.53

Hit@k: at least one relevant item in the top k; Recall@k: fraction of relevant items in the top k; MRR@3: reciprocal rank of the first relevant item within the top 3. Evaluation uses 802 topic-test and 912 random-test queries with available relevance judgments. The random test contains 913 answerable questions in total.

Finding at least one reference-relevant item is easier than recovering every relevant item: dense topic-test Hit@3 is 74.19, whereas Recall@3 is 65.12. Neither measure establishes that the resulting generator input contains all evidence needed to answer, particularly when multiple sources or time-sensitive conditions matter.

6.2 Answerable-Only Generation

Table 3 reports generation metrics on the 802 answerable topic-test questions. mBART benefits from the evaluated retrieval configurations: its ROUGE-L rises

from 12.62 for Q-only to 16.26 for Dense, and EM rises from 1.37 to 2.41. The low absolute EM nevertheless cautions against treating this improvement as sufficient evidence of deployment readiness. The effect is not consistent across models.

Table 3. Topic-split answerable-only results: mean ± SD over three seeds.

Model	Input	EM	Token-F1	ROUGE-L	FR
ViT5-base	Q-only	0.00 ± 0.00	9.44 ± 0.14	8.92 ± 0.08	86.03
ViT5-base	BM25	0.00 ± 0.00	7.55 ± 0.24	7.01 ± 0.17	98.84
ViT5-base	Dense	0.00 ± 0.00	7.78 ± 0.34	7.19 ± 0.24	97.22
ViT5-base	Hybrid	0.00 ± 0.00	7.76 ± 0.36	7.17 ± 0.25	97.46
BARTpho-base	Q-only	2.08 ± 0.52	15.38 ± 0.59	14.76 ± 0.49	52.62
BARTpho-base	BM25	0.46 ± 0.19	15.60 ± 0.49	13.81 ± 0.44	41.06
BARTpho-base	Dense	0.54 ± 0.26	16.98 ± 0.63	14.84 ± 0.52	34.91
BARTpho-base	Hybrid	0.50 ± 0.25	16.72 ± 0.54	14.62 ± 0.40	37.61
mBART-50	Q-only	1.37 ± 0.00	13.45 ± 0.43	12.62 ± 0.43	63.84
mBART-50	BM25	2.41 ± 0.19	18.24 ± 0.59	15.73 ± 0.44	29.05
mBART-50	Dense	2.41 ± 0.31	18.88 ± 0.39	16.26 ± 0.34	30.09
mBART-50	Hybrid	2.37 ± 0.25	18.82 ± 0.47	16.17 ± 0.41	29.38
mT5-base	Q-only	0.00 ± 0.00	3.18 ± 0.95	3.25 ± 0.47	0.00
mT5-base	BM25	0.00 ± 0.00	7.96 ± 0.31	7.72 ± 0.41	0.00
mT5-base	Dense	0.00 ± 0.00	8.16 ± 0.51	7.96 ± 0.58	0.00
mT5-base	Hybrid	0.00 ± 0.00	8.07 ± 0.42	7.83 ± 0.48	0.00
mT5-small	Q-only	0.00 ± 0.00	3.14 ± 0.46	3.17 ± 0.48	0.00
mT5-small	BM25	0.00 ± 0.00	8.98 ± 0.54	8.97 ± 0.40	0.00
mT5-small	Dense	0.00 ± 0.00	8.97 ± 0.46	8.92 ± 0.35	0.00
mT5-small	Hybrid	0.00 ± 0.00	9.14 ± 0.51	9.07 ± 0.41	0.00

All values are percentages, computed on the 802 answerable topic-test questions. EM: exact match after normalization; R-L: ROUGE-L F-measure; FR is the mean fraction of answerable questions labeled as rejection by the output-category parser. For mT5, FR and EM are affected by sentinel-token outputs (Section 5.3).

ViT5's answerable ROUGE-L decreases from 8.92 to 7.01 under BM25, while its false-rejection rate rises from 86.03% to 98.84%. BARTpho exhibits a different trade-off: Dense yields ROUGE-L of 14.84 versus 14.76 for Q-only, but EM decreases from 2.08 to 0.54. A statement that context uniformly improves answer quality would therefore be unsupported by the current results. Figure 1 compares answerable ROUGE-L across both protocols. It illustrates that changing the split or retriever does not produce a uniform model ranking. Because the test sets differ, the curves should be read within each panel before comparing panels. The mBART gain is also protocol-dependent. On the topic test, Dense exceeds Q-only in answerable ROUGE-L in every seed, by 2.88–4.38 points. On the random test, Q-only exceeds Dense in every seed, by 3.19–4.46 points (18.85 versus 15.12 on average). The evidence therefore supports a retrieval benefit for mBART under the topic-disjoint protocol only, not a robust benefit.

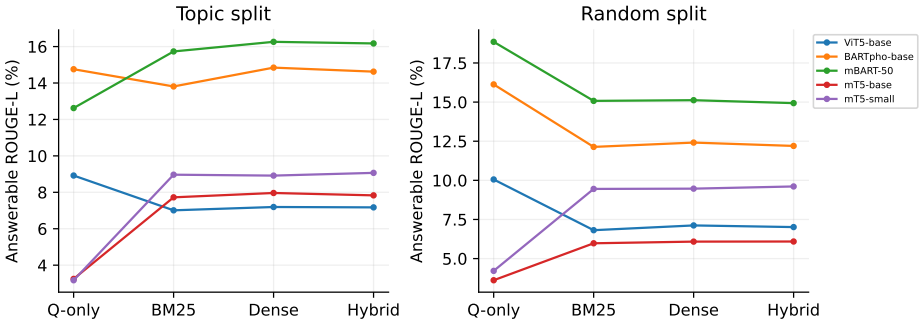


Fig. 1. Answerable-only ROUGE-L means over three seeds. The two panels use different test questions. Topic-split values with standard deviations are given in Table 3; random-split values are included in the repository.

6.3 Rejection Behavior and Metric Interpretation

Table 4 distinguishes binary rejection from recognizing the OOS subtype. A high rejection recall can be obtained by rejecting most questions, including those that can be answered. Consequently, precision and the answerable false-rejection rate must accompany recall. The deterministic always-insufficient baseline reaches 34.89% overall EM without answering any answerable question, demonstrating why aggregate EM cannot serve as the sole system ranking.

No evaluated configuration exceeds the always-insufficient baseline on binary rejection F1 (66.05 for ViT5 Hybrid versus 66.22). Binary F1 alone therefore does not show that any model has learned useful abstention. What separates the models from the baseline is their behavior on answerable questions, where the baseline scores zero: mBART Dense, for example, has lower binary F1 (54.90) but higher binary precision (61.72 versus 49.50) and answers 69.91% of answerable questions. Under every retrieval condition, all models obtain an OOS F1 of 0.00; only Q-only configurations produce any correct OOS outputs. Because 94.40% of topic-test OOS examples come from a single held-out topic, OOS F1 here measures recognition of that topic's questions rather than scope awareness across diverse out-of-scope requests.

The reported FR is defined against corpus-level answerability. A question may be answerable from the full institutional collection while the retrieved input lacks sufficient evidence. In that case, rejecting can be appropriate relative to the input but still count as a false rejection against the corpus label. Separating these interpretations would require judgments of evidence sufficiency for the actual retrieved context. The insufficient-information label thus measures evidence sufficiency at the corpus level, not the sufficiency of the top-3 chunks given to the generator.

Table 4. Rejection detection on the topic test: mean scores over three seeds (percent).

Model	Input	Binary P	Binary R	Binary F1	OOS F1
ViT5-base	Q-only	50.43	89.31	64.46	3.82
ViT5-base	BM25	49.50	98.85	65.97	0.00
ViT5-base	Dense	49.67	97.88	65.89	0.00
ViT5-base	Hybrid	49.72	98.35	66.05	0.00
BARTpho-base	Q-only	58.54	75.74	66.00	1.93
BARTpho-base	BM25	57.89	57.55	57.62	0.00
BARTpho-base	Dense	60.88	55.30	57.85	0.00
BARTpho-base	Hybrid	59.45	56.15	57.63	0.00
mBART-50	Q-only	55.17	80.15	65.34	13.46
mBART-50	BM25	61.60	47.50	53.63	0.00
mBART-50	Dense	61.72	49.45	54.90	0.00
mBART-50	Hybrid	62.46	49.79	55.40	0.00
mT5-base	Q-only	0.00	0.00	0.00	0.00
mT5-base	BM25	0.00	0.00	0.00	0.00
mT5-base	Dense	0.00	0.00	0.00	0.00
mT5-base	Hybrid	0.00	0.00	0.00	0.00
mT5-small	Q-only	0.00	0.00	0.00	0.00
mT5-small	BM25	0.00	0.00	0.00	0.00
mT5-small	Dense	0.00	0.00	0.00	0.00
mT5-small	Hybrid	0.00	0.00	0.00	0.00
Always insufficient	–	49.50	100.00	66.22	0.00
Always OOS	–	49.50	100.00	66.22	25.49

Binary P, R, and F1 treat insufficient and OOS together as the positive class; OOS F1 treats OOS alone as positive. Categories are assigned by the output-category parser (Section 5.3); mT5 zeros reflect the sentinel artifact, not the absence of rejection text.

6.4 Discussion of Five Representative Cases

We inspect five representative examples from the topic-disjoint test under hybrid retrieval with seed 42: three answerable questions, one insufficient-information question, and one OOS question. Complete model outputs for these cases are included in the repository. Cases 1 and 2 illustrate contrasting exact-match outcomes, Case 3 is a randomly selected answerable example, and Cases 4 and 5 illustrate the two rejection categories. These post-hoc examples explain failure modes but do not estimate their prevalence.

Cases 1–3: answerable questions. For a disciplinary-hearing question whose reference answer is "Có.", only mBART matches exactly. BARTpho also begins with "Có." but adds a long procedural account (token-F1=2.50, EM=0), so EM alone would wrongly suggest that its yes/no decision is incorrect. ViT5 rejects the question, and the mT5 outputs contain sentinels with repetitive or uninformative text. For a question on student Gmail accounts, no model matches exactly: mBART keeps the account-allocation statement but omits the required address format (token-F1=26.49), BARTpho and ViT5 reject, and mT5-small

contradicts the reference. For a randomly selected question on email login failure, mBART, BARTpho, and ViT5 all reject the password-reset procedure, showing that an average model ranking does not ensure coverage of individual service requests. Because the retrieved context was not retained for this example, we cannot tell whether retrieval, truncation, evidence use, or rejection tendency caused the failure.

Cases 4–5: rejection questions. For a question on plans to upgrade student-support activities, labeled insufficient information, ViT5 reproduces the correct rejection sentence even though it ranks lowest on answerable token-F1, whereas mBART answers "Có." and BARTpho asserts detailed institutional plans that the benchmark does not support. mT5-base emits only the tail of the rejection template yet obtains token-F1=66.67, an example of substantial lexical overlap without a complete rejection. For a question on a Food Technology master's or exchange program, labeled OOS, BARTpho and ViT5 emit the insufficient-information sentence: they count as binary rejections but do not identify the scope boundary, which the benchmark must make explicit because study abroad is related to university services in ordinary usage.

Implications for evaluation. The cases illustrate over-generation, omission of requested details, over-rejection, unsupported affirmation relative to the benchmark, incomplete rejection language, and confusion between rejection subtypes. They motivate reporting answerable-only quality, binary rejection scores, and subtype scores together, and they caution against treating lexical or embedding similarity as factual correctness. A source-grounded human assessment would be needed to distinguish retrieval failures from unsupported generation and to judge long answers that fail exact matching.

7 Limitations

This study is institution-specific and evaluates one fixed split per protocol. Topic and answerability are associated, and the random partition does not by itself establish resistance to paraphrase leakage. The Q-only/RAG comparison changes both training and inference. The mT5 sentinel issue limits exact-match and rejection interpretation. BERTScore values are high even for some low-overlap predictions; they are retained only as similarity diagnostics and should not be read as correctness probabilities. Inter-annotator agreement and human-edit rates were not measured, and the output-category parser was not validated against human judgments. No gold-context, LLM-generator, or threshold-based abstention baseline is reported, and no source-grounded human or LLM-as-judge assessment of long answers was conducted. Future work should add an oracle-context condition, repeat the topic-disjoint evaluation over several topic partitions to cover more diverse OOS questions, and assess a stratified sample of answers against the source evidence. These limitations motivate transparent component-level reporting rather than a claim of uniformly reliable Student Affairs QA.

8 Conclusion

We evaluated five sequence-to-sequence models with three retrieval configurations on a revised Vietnamese Student Affairs benchmark containing 15,879 samples. Separate retrieval, answerable-only generation, and rejection analyses reveal behavior obscured by aggregate scores. Dense retrieval with reranking performs well on the topic-test retrieval measures, and mBART improves with retrieved context on the topic-disjoint test, though not on the random split, but other models show over-rejection or mixed metric changes. Constant rejection baselines and the mT5 output-format issue further demonstrate why reliable evaluation requires inspecting both the content and the interpretation of generated responses. The present findings support model-dependent benefits and clear remaining limitations, rather than a general claim that adding context consistently improves QA.

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